

202504 - LASSO - Unbounded subproblem case

Timeline in: [Column Generation For Regression Regularization](#)

Overview

1.2 Relaxation subproblems

Furthermore, the relaxed subproblems become **completely independent/trivially separable**, with each one of them solving a subproblem of the form

$$(L_i) \min_{\beta_i} (\psi' X)_i \beta_i + \lambda_1 |\beta_i| + \lambda_2 \beta_i^2,$$

which has a closed-form solution, depending on the sign of $(\psi' X)_i$, as

- If $\lambda_1 < |(\psi' X)_i|$, then $\beta_i = \frac{-(\psi' X)_i + \text{sign}[(\psi' X)_i] \cdot \lambda_1}{2\lambda_2}$.
- If $\lambda_1 \geq |(\psi' X)_i|$, then $\beta_i = 0$.

1.2.1 LASSO and Ridge regression extension.

The condition for those two problems is exactly the same, but setting respective opposite $\lambda_i = 0$. In other words:

1. **LASSO:** We set $\lambda_2 = 0$, to get
 - If $\lambda_1 < |(\psi' X)_i|$, then $\beta_i \rightarrow -\text{sign}[(\psi' X)_i] \cdot \infty$ (**unbounded subproblem**).
 - If $\lambda_1 \geq |(\psi' X)_i|$, then $\beta_i = 0$.
2. **Ridge:** We set $\lambda_1 = 0$, to get (a **single condition**)
 - If $0 < |(\psi' X)_i|$, then $\beta_i = \frac{-(\psi' X)_i}{2\lambda_2}$.
 - If $0 = |(\psi' X)_i|$, then $\beta_i = 0$.

⚠ Note: We also need to verify if it is worth to use SOC formulations for β_i^2 terms on the master problem or not, in terms of performance.

1.2.1.1 LASSO: Unbounded subproblem case

Let us continue this section on a new notebook: [202504 - LASSO - Unbounded subproblem case](#).

0. What is the theoretical behavior of the bound?

I am assuming constraining the master problem to just a set of columns is somewhat an equivalent idea to having an integer feasible region on the master problem, like the [Lagrangian methods Lecture](#) of IP, where they get the relationship $z_{LO} \leq z_D \leq z_{IO}$, where IO is the integer problem, D is the dual problem obtained in the classical dual max but with integrality of IO (subregion of columns in our case), and LO is the linear relaxation (full columns problem in our case).

But is not the same in our case, since we have that the master problem gives us an upper bound (subregion because of subset of columns), and the Lagrangian relaxation gives us a lower bound, which also makes sense since we are relaxing a constraint, and just adding some pricing on it, but the subregion is still bigger than the original problem.

1. Unbounded LASSO subproblem

We have that the LASSO subproblem is unbounded, i.e., with $\lambda_1 > 0$ the problem

$$(L_i) \min_{\beta_i} (\psi' X)_i \beta_i + \lambda_1 |\beta_i|$$

becomes unbounded due to the condition $|(\psi' X)_i| > \lambda_1$, because the regularization term does not have sufficient penalty. The objective function is just a linear function plus absolute value, both convex function, and thus this is an unconstrained convex optimization problem.

One could think first about the First-Order necessary & sufficient condition for this case, but this function does not have any saddle point if the problem is unbounded, and when bounded the solution is obvious, $\beta_i = 0$.

1.1. Unit ball constraint

If $\exists i \in [m] : |(\psi' X)_i| > \lambda_1$, then at least one subproblem is unbounded, then we do **not have any useful solution** to add to the master problem. Therefore, instead of thinking about producing a proper solution, one could instead think of producing a useful extreme ray to add to the MP. Then, the natural alternative would be to solve the complete subproblem with a **bounding constraint, like the unit ball**. In other words, we could solve:

$$(L) \min_{\beta} \quad \lambda_1 \|\beta\|_1 + \psi' X \beta - \psi' y - \frac{\mu^2}{4} \\ s. t. \quad \|\beta\|_2 \leq 1,$$

Since the dual variables already have the **information about the master problem solution** (similar to adding x of master problem to the subproblem, then this should lead to a direction of steepest descent, or a similar idea, on the unbounded case [**check theory behind this**]).

1.2. Adding steepest descent + bounded solution

Again, if $\exists i \in [m] : |(\psi'X)_i| > \lambda_1$, then least one subproblem is unbounded. BUT, we still have the information about the primal solution of the master problem $\beta_{MP} = \sum_{k=1}^K \pi_k \beta^k$. Using this solution, we can compute the **direction of steepest descent of the Lagrangian** from a given β , i.e. $d = -\nabla L(\beta) = -(\psi'X)_i - \lambda_1 \frac{\beta_i}{|\beta_i|}$. Thus, we can compute the direction of steepest descent from any solution, for example β_{MP} and **feed it to the next master problem**, to get a next problem of the form $\pi\beta_{MP} + (1 - \pi)d$, which adds information in a similar way as in the idea behind gradient descent, i.e., $\beta_{MP_{t+1}} = \pi\beta_{MP_t} - (1 - \pi)\nabla L(\beta_{MP_t})$.

1.2.1. Estimation of the gradient.

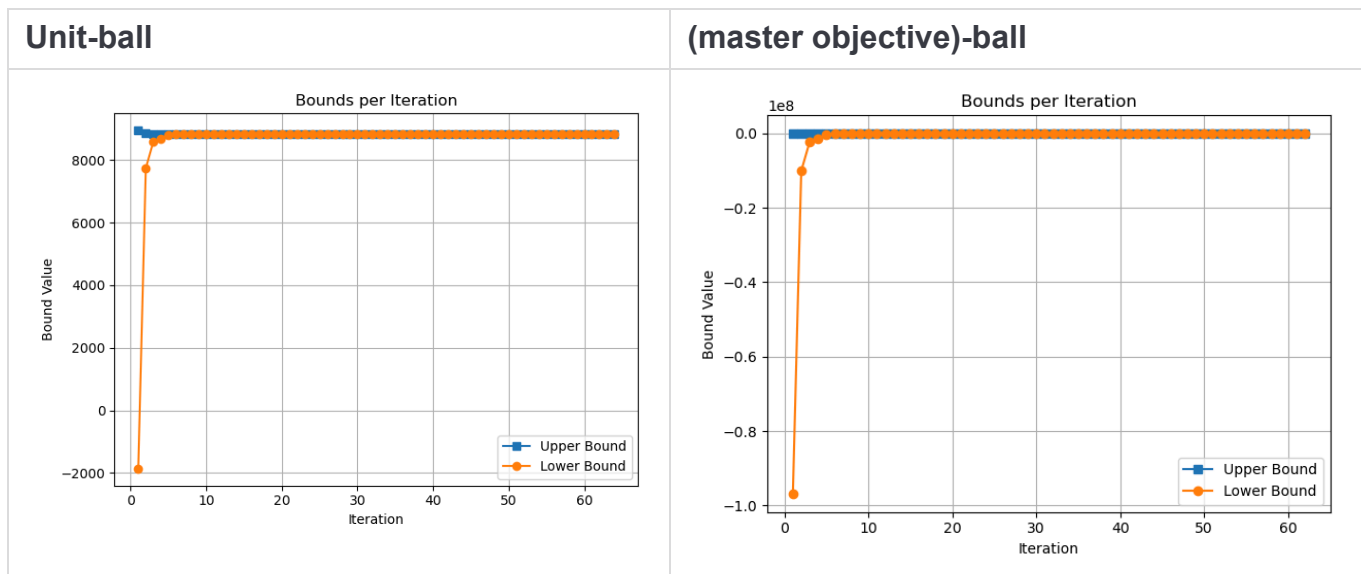
vi

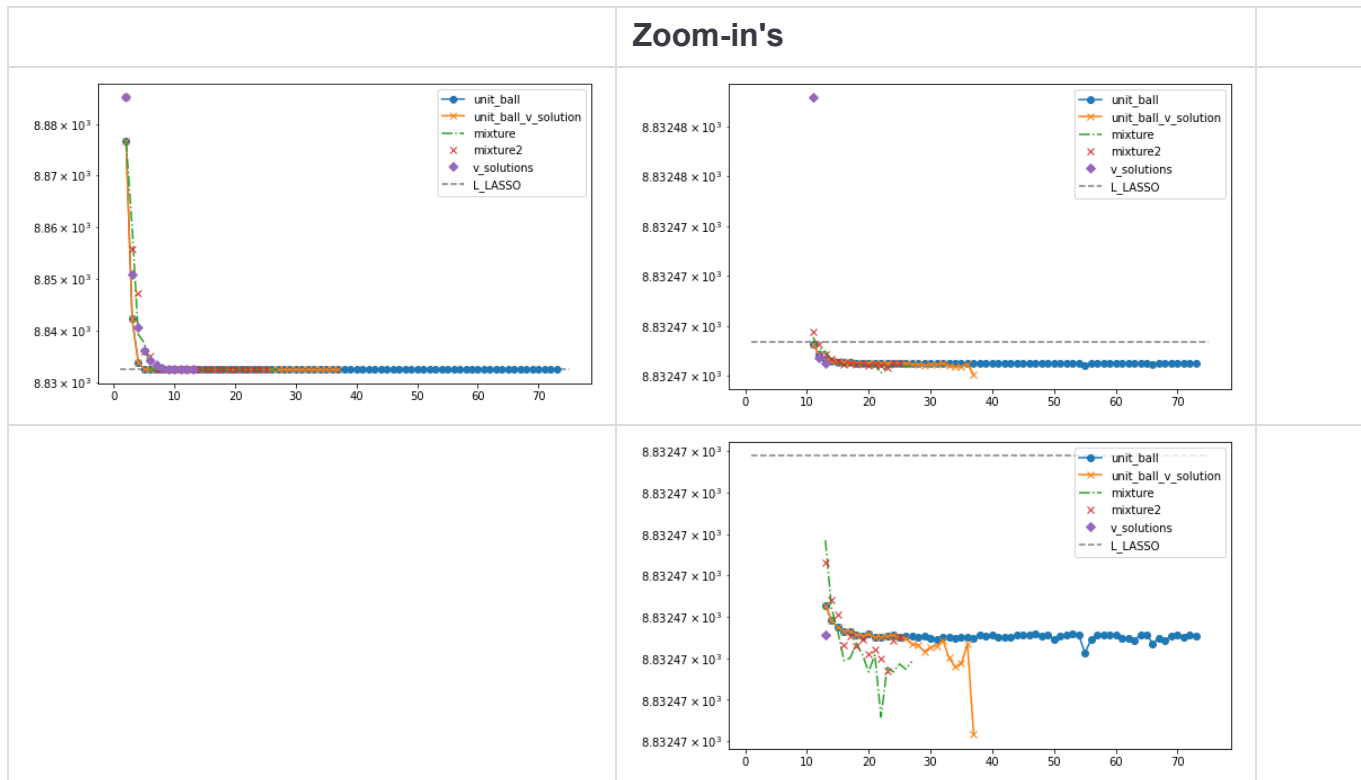
Results

1.1 Unit ball constraint

We can get a better behavior of the lower bound (Lagrangian) and upper bound (master problem), in the sense that they get bounds that get closer very quickly, and in the old version **we didn't even have a lower bound**. The problem is that it fails to converge when closer to the upper bound, and starts to jump high and low (the lower bound).

Our approach makes sense in theory then, but





Notes:

- The unit ball method (analogous behavior as minus gradient) have a faster approach to the optimal value, but then it starts to oscillate instead of converge.
- I tried to combine them as first going with good direction of unit ball, and then use $v_solutions$, but it did not work quite well, but at least better than unit ball alone.
- It improved more doing mixtures of 1 iter of unit_ball and 1 iter of $v_solutions$, but still worse than $v_solution$ alone.
- Maybe analyze what happens to the solution itself when close to the optimal value, as what and how many coordinates are active/zero/etc.

1.2 Computing the direction exactly from $\phi'X$

1.3 SGD-like computation

Using subsamples of $\phi'X$: X is too big, so sample different mini-batches (usually of bit-multiple sizes like: 32, 64, etc.) from the observations of X (rows), and compute multiple divergent directions.

If exact computation is expensive, replace it with a cheaper estimate

Pro: Saves computing $\phi'X$, and probably gives a better estimate of the divergent direction than just $\phi'X$ in most of the cases (does not overfit to it).

1.4 Current approach: Compute only the most important (largest) v components.

Focusing on keeping only the v largest coordinates of divergence.

Future Note

Do a profiling of the method, to see what are the most time-consuming operations, and analyze how to improve them.

ITERATION 58

Master Problem

The optimal value is 8876.560641378746

Time elapsed in iteration 58

0.02829245726267497 mins (normal)

0.05859375 mins (process)

First stopping criterion NOT met: $\lambda_k \neq \lambda_{k-1}$

Skipping Lagrangian Model

Lagrangian is -infinity

beta_size 1000

m 1000

beta_sol shape: (1000,)

shape of X in fit (1000, 69)

shape of y in fit (1000,)

Second stopping criterion NOT met: $\text{residuals}^2 \neq 0$

Time limit reached

Plotting bounds...

Tags

[#Tesis](#) [#Uchile](#) [#ColumnGeneration](#)